SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	M. Sneha	**Reg. No.:**	192425092
Section / Batch:	AI&ML	**Date:**	

Code of Conduct

I ________M. Sneha(192425092)_________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _______M. Sneha________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Answers

1. Frame Semantics in Banking Customer-Service Systems – Answers

Task 1: Semantic frame relationships

Each customer query is mapped to a predicate–argument (frame) structure in which the verb defines the action and the surrounding noun phrases are bound to fixed roles. TRANSFER(SourceAccount, Amount, DestinationAccount, Customer) treats the Customer as Agent, the Amount as Theme, and the two accounts as Source and Destination. BLOCK(DebitCard, Customer) binds Customer as Agent and DebitCard as the affected object (Patient). SEND(TransactionStatement, Email, Customer) treats the Customer as Agent, the statement as Theme, and the email address as Goal/Destination. MODIFY(WithdrawalLimit, Customer) binds Customer as Agent and WithdrawalLimit as Theme, but — critically — does not encode the direction of the modification (increase or decrease) as a separate argument.

Task 2: Query with incorrect interpretation

Query B4 is incorrect. The actual user intent was to increase the withdrawal limit, but the system interpretation records a decrease. Since both B1–B3 show the system frame matching the true intent, the error is isolated to B4, where the MODIFY frame captured the correct entity (WithdrawalLimit) but failed to correctly resolve the directional/polarity argument (increase vs. decrease), effectively inverting the customer's request.

Task 3: Effect of incorrect semantic-role identification

When a semantic role (such as direction, source, or destination) is wrongly bound, the system executes an action opposite to or different from what the customer intended. In B4 this means the customer's daily withdrawal limit is lowered instead of raised, which can block legitimate transactions and cause financial inconvenience. More generally, if Source and Destination were swapped in a TRANSFER frame, money could be pulled from the wrong account; if the wrong Patient were bound in a BLOCK frame, the wrong card could be blocked. Because these frames drive real financial operations, an incorrect role assignment translates directly into an incorrect and potentially harmful banking action, with consequences for customer trust and regulatory compliance.

Task 4: Recommended improvement

A more reliable frame-based approach should:

• Make polarity/direction an explicit, mandatory slot in frames such as MODIFY (e.g., MODIFY(WithdrawalLimit, Direction, Customer)) so it cannot be silently dropped or defaulted.

• Use dependency parsing to correctly attach modifier words ("increase", "decrease", "raise", "lower") to the correct frame argument rather than relying on keyword matching alone.

• Add a confirmation step for high-impact operations (fund transfers, limit changes) where the extracted frame is echoed back to the customer before execution.

• Introduce confidence scoring for frame extraction, routing low-confidence cases to human review or clarifying questions.

• Continuously validate frame outputs against business rules (e.g., plausible transaction ranges) to catch logically inconsistent role bindings before they are executed.

2. First-Order Predicate Calculus for Smart Agriculture – Answers

Task 1: Field observations in FOPC

F1: $Dry(F1) \land IrrigationAvailable(F1) \land Rice(F1)$

F2: $Wet(F2) \land IrrigationAvailable(F2) \land Wheat(F2)$

F3: Dry(F3) ∧ IrrigationUnavailable(F3) ∧ Maize(F3)

F4: Wet(F4) ∧ IrrigationUnavailable(F4) ∧ Rice(F4)

Task 2: Fields requiring irrigation

Applying the rules: for F1, Dry(F1) ∧ IrrigationAvailable(F1) → NeedsWater(F1), and NeedsWater(F1) → Irrigate(F1); since Rice(F1) ∧ NeedsWater(F1) also holds, PriorityIrrigation(F1) is derived. For F2, Wet(F2) → ¬NeedsWater(F2), so no irrigation is triggered. For F3, the rule Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x) does not fire because IrrigationAvailable(F3) is false; instead IrrigationUnavailable(F3) → ¬Irrigate(F3) applies directly, so F3 is not irrigated even though it is dry. For F4, Wet(F4) → ¬NeedsWater(F4) and IrrigationUnavailable(F4) → ¬Irrigate(F4) both hold, so F4 is not irrigated. Conclusion: only F1 requires and receives irrigation, and it is additionally flagged as a priority field because it grows rice.

Task 3: Should F1 and F3 be treated differently?

Yes. F1 is dry and has irrigation available, so the rule base cleanly derives NeedsWater, Irrigate, and PriorityIrrigation — the controller can safely automate irrigation. F3 is also dry (i.e., physiologically in need of water) but irrigation is unavailable; the rule that would derive NeedsWater(F3) never fires because it requires availability, so the system only concludes ¬Irrigate(F3) from the resource constraint, not that the field has no need for water. Treating F1 and F3 identically would be wrong: F1 can be left to automated irrigation, whereas F3 should be flagged for manual intervention — e.g., alternate water sourcing, rescheduling, or priority restoration of irrigation supply — because the logic base never actually confirms F3 does not need water; it merely reflects that the resource is currently unavailable.

Task 4: Sufficiency of predicate logic alone

Predicate logic alone is not sufficient for real agricultural decision support. First-order predicate calculus is monotonic and strictly binary (true/false), so it cannot represent degrees of soil dryness, uncertainty in sensor readings, probability of sensor failure, or partial/missing data — all common in field deployments. It also has no built-in mechanism for weighing competing priorities (e.g., two dry fields needing water but only one irrigation pump available) or for temporal reasoning about how long a field has been dry. For robust decision support, the rule base should be supplemented with probabilistic or fuzzy reasoning (e.g., Bayesian networks or fuzzy soil-moisture degrees) to handle uncertain and incomplete sensor data, and with an optimization/scheduling layer to allocate limited irrigation resources across competing needs.

3. Word Sense Disambiguation in a Travel Recommendation System – Answers

Task 1: Intended sense using context

"Amazon tour", followed by viewing rainforest trekking packages, resolves to the Amazon rainforest sense. "Safari booking", followed by selecting wildlife resort packages, resolves to the wildlife-tour sense. "Java vacation", followed by viewing hotels in Bali and Java, resolves to the Java island sense — the co-occurring destination "Bali" anchors the geographic reading. "Cruise package", followed by viewing Mediterranean ship packages, resolves to the ship-journey sense. In contrast, "Safari download for Windows", followed by a click on a browser download page, resolves "Safari" to the web-browser sense — the opposite meaning of the same word seen in "Safari booking".

Task 2: Lexical and contextual cues

• Collocated words: "tour", "vacation", "package", "booking" bias toward travel senses, while "download" and "for Windows" bias toward the software/browser sense.

• Co-occurring place names (e.g., "Bali" alongside "Java") confirm a geographic rather than technical sense.

• Platform/technology terms ("Windows") signal an operating-system or software context.

• Post-search behaviour (which result the user clicks) is a strong implicit signal that confirms or overrides the lexical cues.

Task 3: Why the same word needs different interpretations for different users

Word meaning in natural language is context- and intent-dependent rather than fixed per token: "Safari", "Java", "Amazon", and "Cruise" are each polysemous, and the correct sense depends on the surrounding query terms and on what the individual user is actually trying to accomplish. One user searching "Safari" is planning a wildlife trip, while another is trying to install a browser — the token is identical, but the discourse context and user goal differ, so the system must disambiguate per-session rather than assume a single global sense for the word.

Task 4: Industrial-scale WSD strategy

• Combine local query-context features (surrounding words) with contextual embeddings (e.g., transformer-based sense representations) that capture sentence-level meaning rather than a single static vector per word.

• Incorporate user history and session context to bias sense priors toward a user's demonstrated interests (e.g., prior travel searches vs. prior software searches).

• Use click-through and post-search behaviour as an implicit feedback signal to continuously retrain and calibrate the sense classifier.

• Ground candidate senses in a knowledge base or domain ontology (e.g., a travel taxonomy versus a software taxonomy) to constrain the sense inventory.

• Apply confidence thresholds, falling back to blended or clarifying results when disambiguation confidence is low.

• Continuously evaluate and retrain the model with online/A-B testing so the system adapts as query patterns and user behaviour evolve.

4. Syntax-Driven Semantic Analysis in Legal Document Processing – Answers

Task 1: Influence of syntactic structure on semantic-role assignment

In active-voice Subject–Verb–Object sentences, the grammatical subject typically coincides with the semantic Agent and the object with the Theme/Patient — e.g., in "The supplier delivered the equipment," Supplier (subject) is the Agent and Equipment (object) is the Theme. In passive constructions, this correspondence breaks: the surface subject is demoted from the syntactic Agent position. In "The contract was terminated by the company," the grammatical subject "Contract" is semantically the Theme/Patient (the thing being terminated), while the true Agent, "company," appears in the by-phrase. Correct semantic-role assignment therefore requires detecting voice (active vs. passive) and using grammatical relations such as the by-phrase, rather than assuming subject position always equals Agent.

Task 2: Incorrect semantic-role assignments

The fourth interpretation — "The contract was terminated by the company" → Contract = Agent, Company = Object — is incorrect. Because the sentence is passive, the correct roles are Company = Agent (performer of the termination, from the by-phrase) and Contract = Theme/Patient (the entity being terminated). The system has mistakenly mapped the grammatical subject directly to the semantic Agent without accounting for the passive voice transformation.

Task 3: Effect of confusing grammatical subject with semantic agent

In legal information extraction, mixing up grammatical subject and semantic agent can invert responsibility and obligation: an automated system might conclude that the contract terminated the company, rather than the company terminating the contract. In practice this can misattribute breach of contract, liability, or performance obligations between parties, leading to incorrect compliance findings, faulty dispute-resolution recommendations, or flawed legal analytics that could mislead downstream decision-making.

Task 4: Recommended syntax-driven semantic analysis architecture

• Use a full dependency parser to extract grammatical relations (nsubj, dobj, nsubjpass, and agent/by-phrase) instead of simple Subject–Verb–Object pattern matching.

• Add an explicit voice-detection step (active vs. passive) before role assignment, applying voice-specific mapping rules or a semantic role labelling (SRL) model that assigns roles such as ARG0 (Agent) and ARG1 (Patient/Theme) independent of surface word order.

• Handle prepositional phrases explicitly ("by the company" marks Agent; "to the buyer"/"from the supplier" mark Recipient/Source) rather than ignoring them.

• Use a neural SRL model trained/fine-tuned on legal text to correctly resolve long-distance dependencies and embedded or coordinated clauses common in contracts.

• Incorporate coreference resolution so that pronouns and repeated references across a long legal sentence or paragraph are correctly linked to the right entities.

• Validate the resulting role assignments against a legal domain ontology or rule base to catch implausible bindings (e.g., a contract acting as an Agent) before the extraction is finalized.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	5	5	5	5	5	25	87
2	5	5	4	4	4	22	
3	4	4	4	4	4	20	
4	4	4	4	4	4	20	

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________